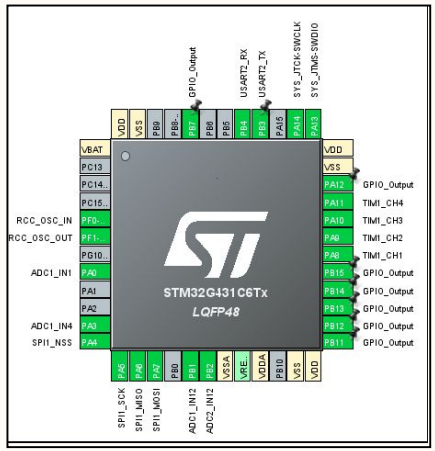
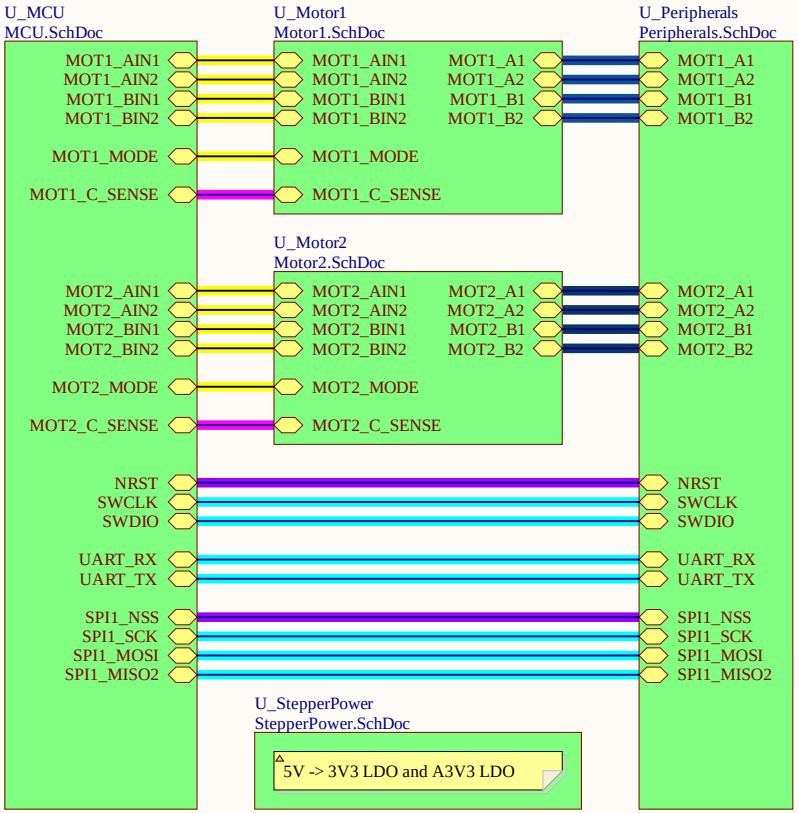
UQ Biorobot New Tracker
Stepper Driver Module

System Overview - All Sheets

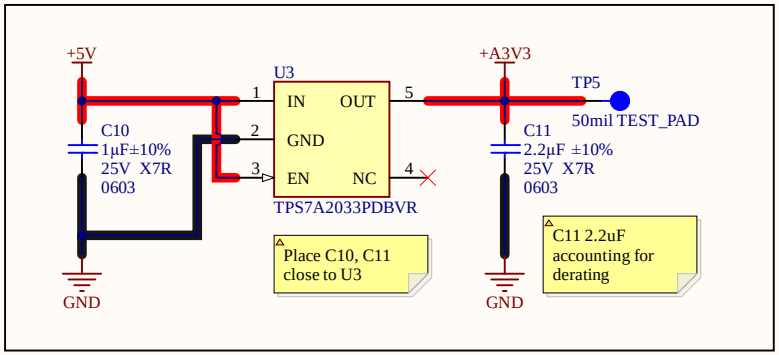
This board allows for control of two bipolar stepper motors for the motorized lens assembly: Zoom/Focus. Each motor has two phases and is driven by a DRV8835 Stepper driver. In addition, in order to sense the end of gear travel, and automatically calibrate the position of the motor, high side current sensing is implemented, measuring the current draw to each motor / motor driver via the INA180A1 (Gain = 20). The entire system is controlled via a STM32G431C6T6 MCU. Peripherals include: SWD, SPI, UART (All JST SH 1.0). A power connector is used to deliver power at 5V 500mA nominal (When both motors are stalled and all phases are powered on). As this board can be powered directly from a Lipo battery, a 1.5A PTC fuse and pmos reverse polarity protection is also applied.



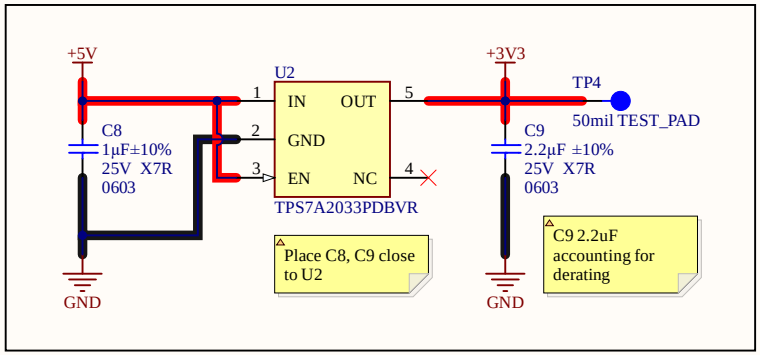


Title: Overview		
Size: A4	Number: 1	Revision: 1
Date: 10/07/2026		Sheet 1 of 6
File: C:\Users\..\Overview.SchDoc		Drawn By: Xuyang Liu

5V to 3.3V LDO (Analog Rail)

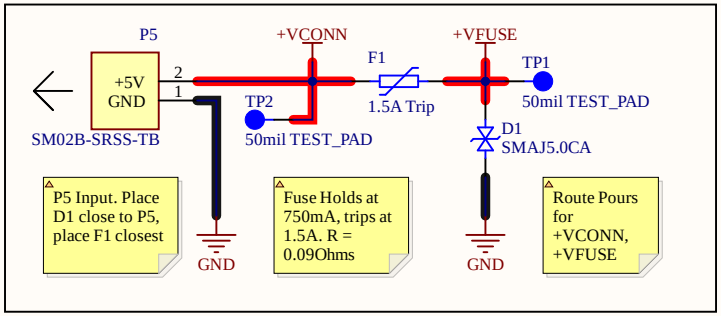


5V to 3.3V LDO

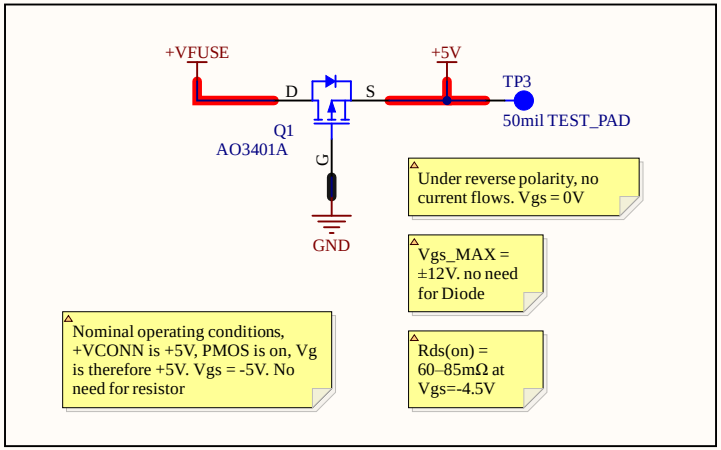


Title: StepperPower		
Size: A4	Number: 2	Revision: 1
Date: 10/07/2026		Sheet 2 of 6
File: C:\Users\..\StepperPower.SchDoc		Drawn By: Xuyang Liu

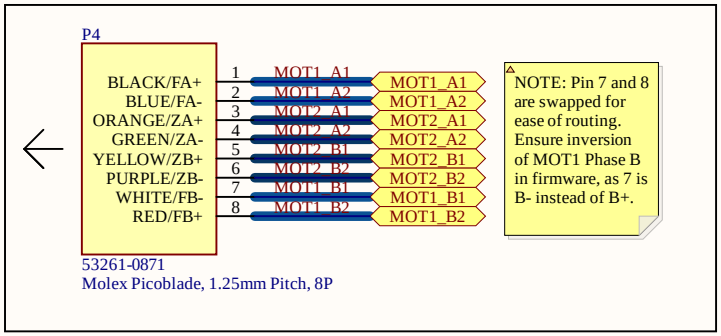
JST SH 1.0 2P +5V Power Connector



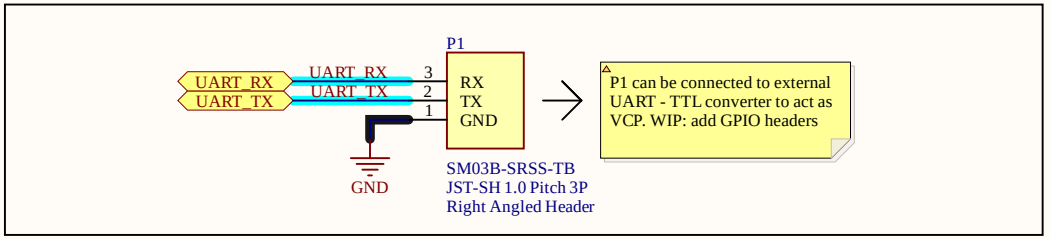
PMOS Reverse Polarity Protection



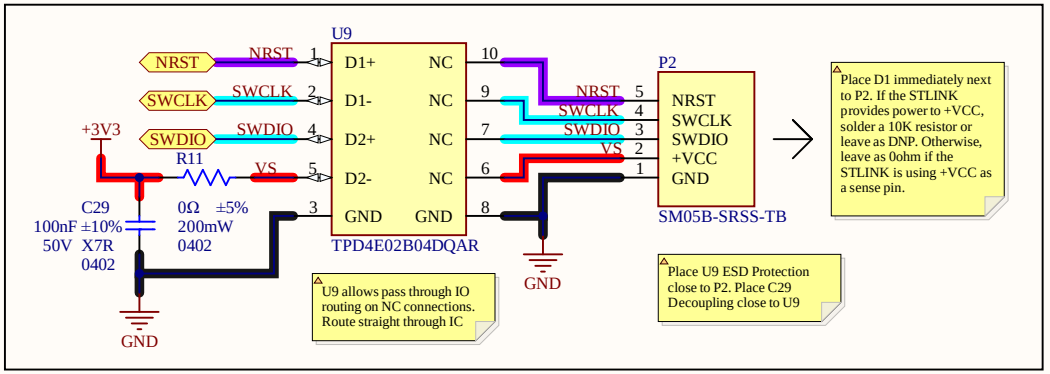
Molex Picoblade Stepper Motor Connector



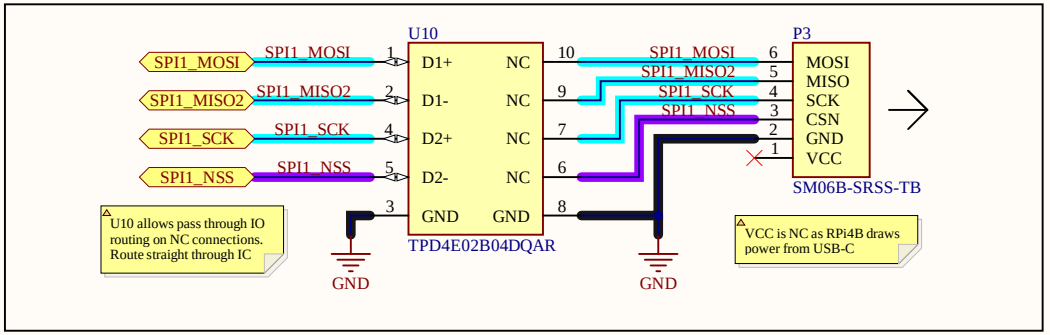
3P UART header



SWD Programming Header

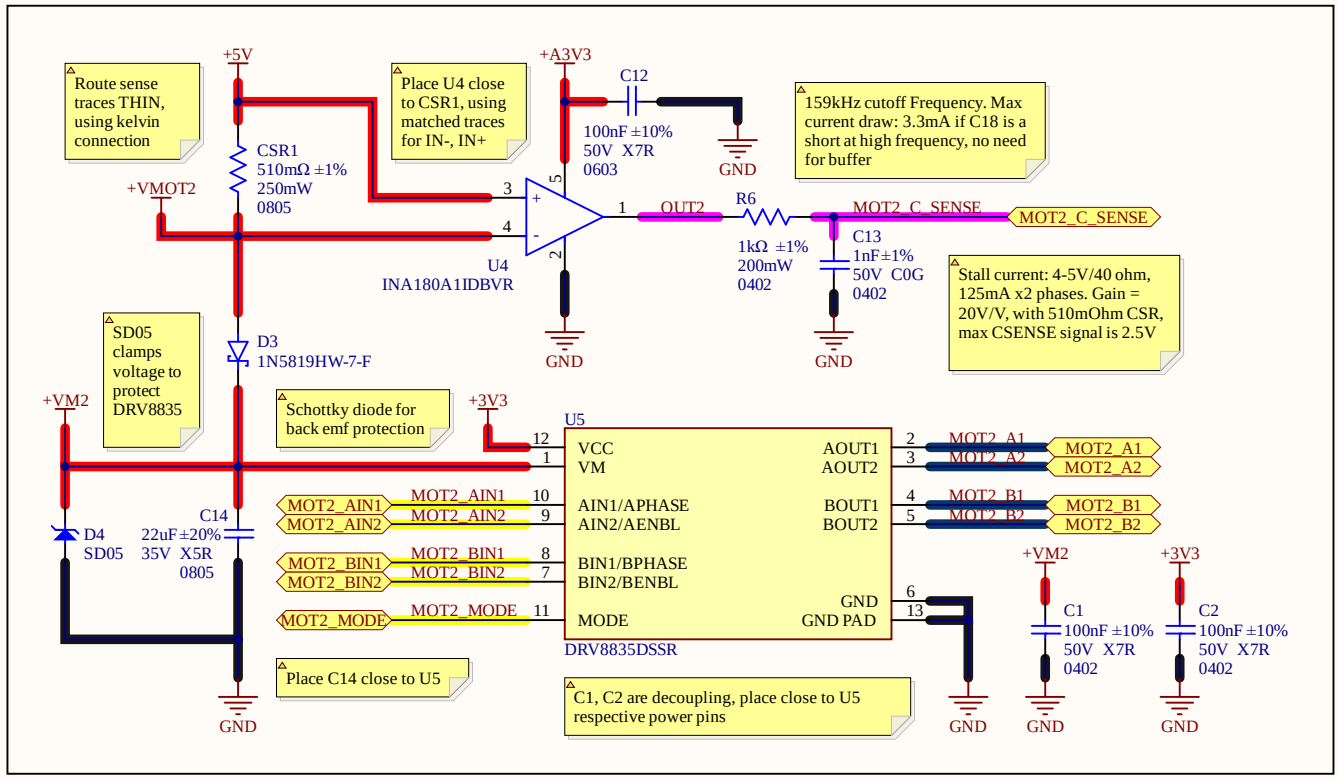


SPI Header to Raspberry Pi 4B



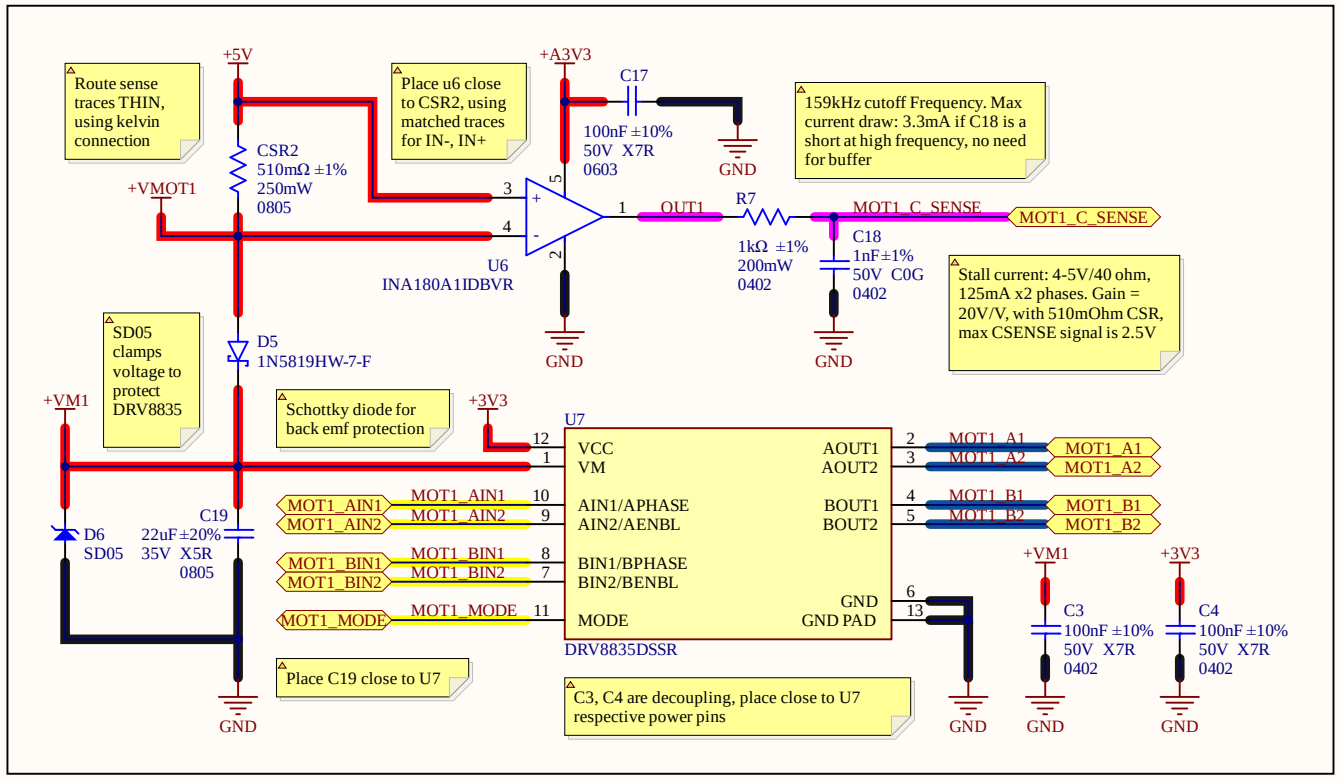
Title: Peripherals		
Size: A4	Number: 3	Revision: 1
Date: 10/07/2026	Sheet 3 of 6	
File: C:\Users\..\Peripherals.SchDoc	Drawn By: Xuyang Liu	

DRV8835 Motor Driver with INA180A1 high side current sensing



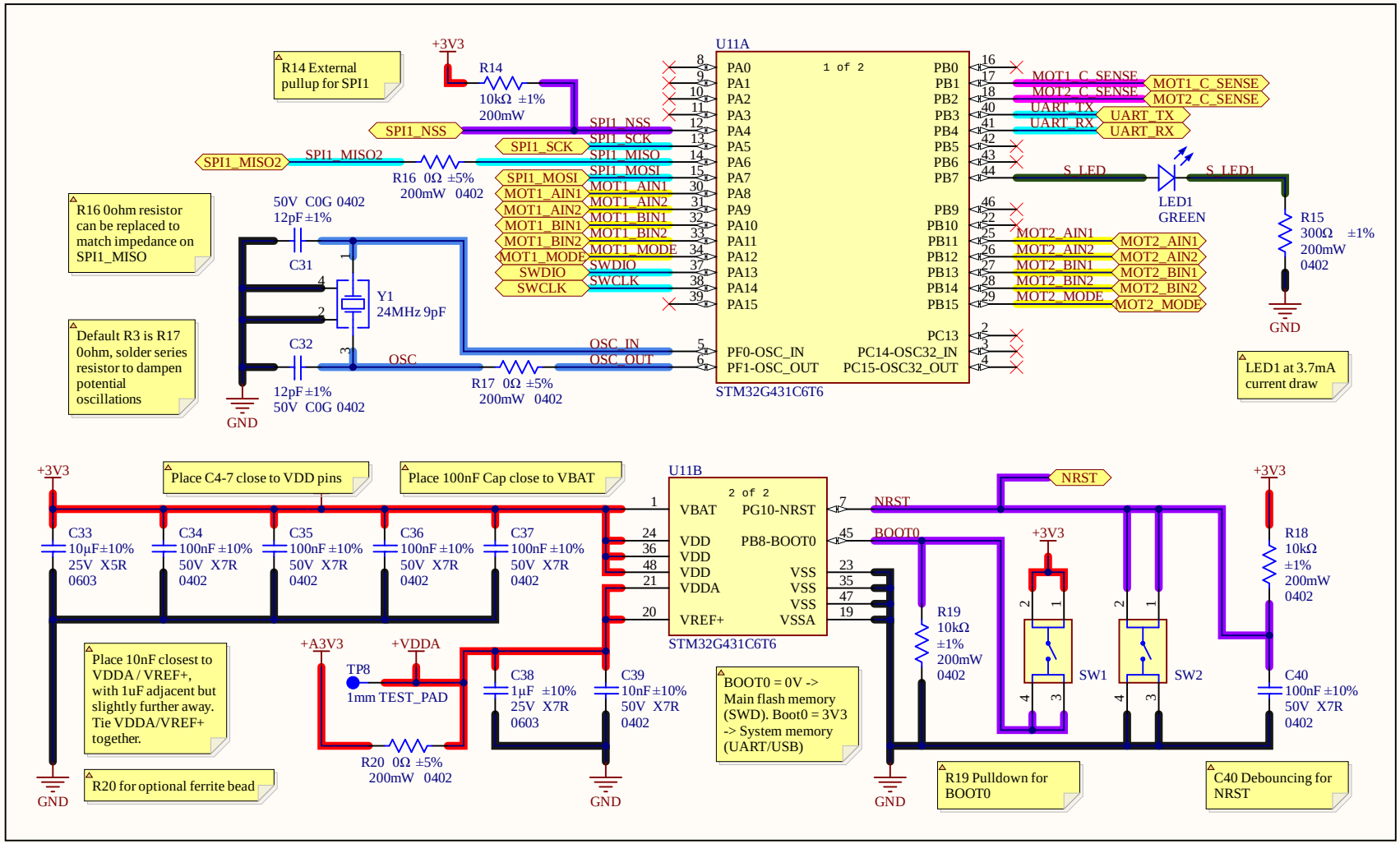
Title: Motor2		
Size: A4	Number: 4	Revision: 1
Date: 10/07/2026	Sheet 4 of 6	
File: C:\Users\..\Motor2.SchDoc	Drawn By: Xuyang Liu	

DRV8835 Motor Driver with INA180A1 high side current sensing

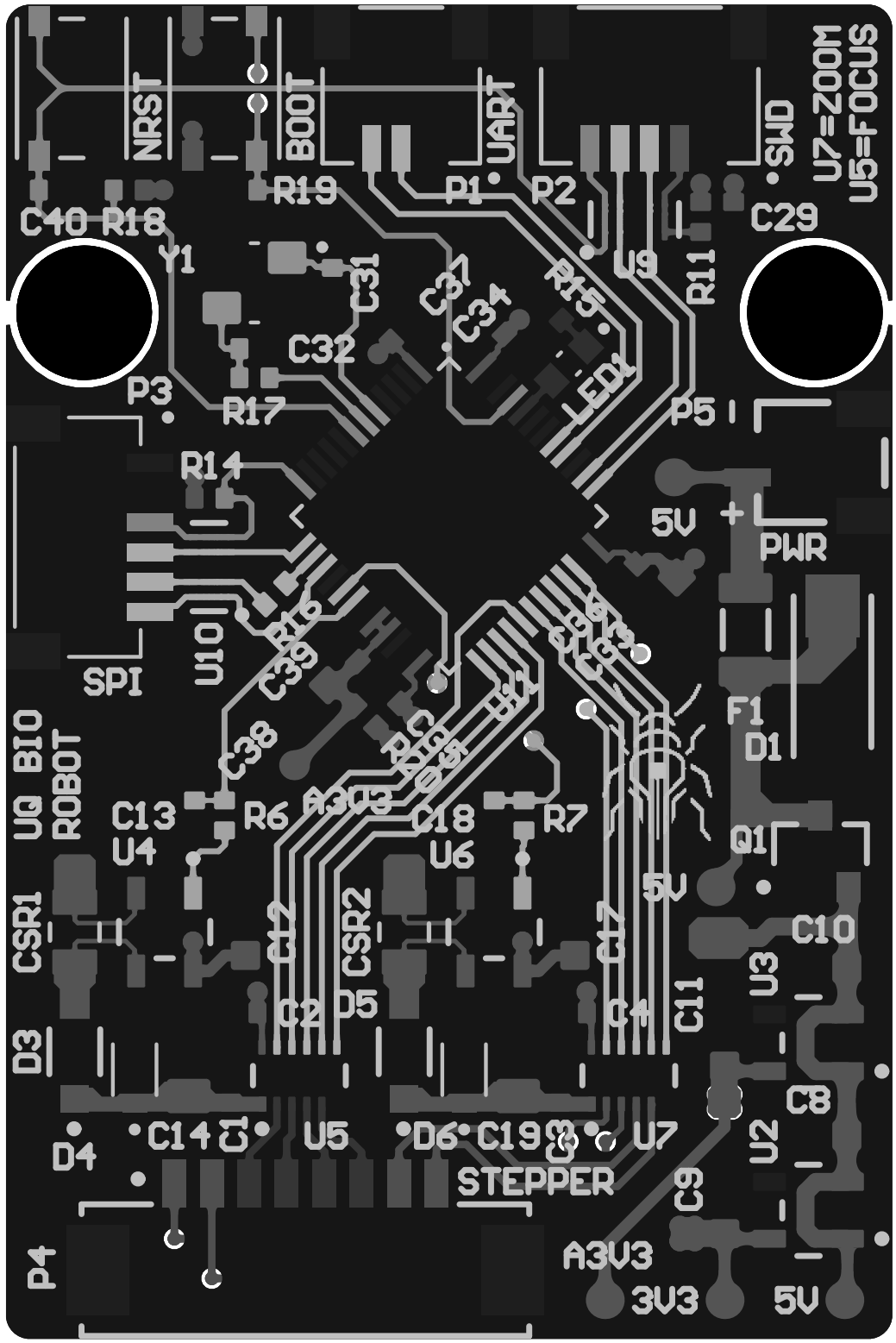


Title: Motor1		
Size: A4	Number: 5	Revision: 1
Date: 10/07/2026	Sheet 5 of 6	
File: C:\Users\..\Motor1.SchDoc	Drawn By: Xuyang Liu	

STM32G431C6T6 Microcontroller



Title: MCU		
Size: A4	Number: 6	Revision: 1
Date: 10/07/2026	Sheet 6 of 6	
File: C:\Users\..\MCU.SchDoc	Drawn By: Xuyang Liu	


NRST
BOOT
UART
SWD
U7=ZOOM
U5=FOCUS
R19
P1
P2
C40
R18
Y1
C31
C37
C34
R15
U9
R11
C29
C32
P3
R17
LED1
P5
R14
5V
PWR
U10
SPI
C39
R16
C38
F1
D1
UQ BIO
ROBOT
C13
U4
R6
A3V3
C18
U6
R7
Q1
CSR1
CSR2
C12
C17
C2
D5
C1
C11
U3
C10
D3
D4
C14
U5
D6
C19
U7
STEPPER
U2
C8
C9
P4
A3V3
3V3
5V